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Editorial

# Nutrition for Healthy Longevity

We shifted our focus from "living longer" to "living better for a longer time", that is, aging healthier. The World Health Organization (WHO) proposes health aging as "developing and maintaining the functional ability that enables wellbeing". Functional ability is the result of interactions between the intrinsic capacity (IC) of an individual and its environment [1]. IC is understood as a person's mental and physical capabilities covered by five subdomains: locomotion, vitality, cognition, psychological, and sensory, which can be maintained for longer with environmental support and prevention/compensation strategies [2]. Nutrition is a core lifestyle factor that has been shown to play a role in all IC domains, thus contributing to health longevity.

Given the importance of muscle for locomotion, protein is usually the first dietary target, especially since older adults, who often present sarcopenia [3], tend to ingest less than their needs. Indeed, Whey protein supplementation (a facilitator to achieve protein recommendations) was shown to increase IGF-1 and albumin while reducing inflammation, resulting in better physical performance [4]. However, eating more protein than current recommendations does not prevent physical function decline over time [5]. Interestingly, diets typically associated with better locomotion include the Mediterranean and the Baltic Sea Diet, in which points are subtracted for some protein-rich foods, especially red meat, while points are awarded for other protein-rich foods like nuts, cereals, legumes, and fish [6]. This indicates that not only the amount of protein but also its source plays a role, as previously suggested for managing frailty [7]. In addition, those dietary patterns provide nutrients with antioxidant and anti-inflammatory properties, such as magnesium (Mg). This mineral prevents cell damage and regulates calcium homeostasis, being essential for muscle function, balance, and bone mass maintenance [8].

Muscle performance usually tracks muscle strength, which reflects neuromuscular function –a component of the Vitality IC domain. Greater grip strength was shown to be associated with higher adherence to Mediterranean [9] and MIND diets (The Mediterranean-DASH Intervention for Neurodegenerative Delay) [10], dietary antioxidants, such as vitamin C, selenium, and carotenoids [11], and dietary protein [12], especially plant-protein [13,14]. On the other hand, the higher the intake of foods from animal sources [15], the lower the grip strength; and the higher the intake of Ultra Processed foods (UP foods) –usually high in refined sugars, fat, and sodium– the faster the grip strength decline [16]. Vitality represents the underlying physiological component of IC, reflecting vulnerability to adverse health outcomes based on reserves and functioning, which can be estimated by assessing appetite and weight loss [17]. Appetite loss during aging (anorexia of aging) is associated with accelerated biological aging [18] and adverse health outcomes, including depressive symptoms [19].

Sugar is another dietary component that deserves attention due to its associations with mental and cognitive disorders. Excessive intake of all sugars and increased intake of some subtypes, such as free sugars, lactose, and sucrose, might interact in ways that exacerbate inflammatory responses, induce synaptic plasticity, and alter the modulation of brain-derived neurotrophic factors (BDNF) and other receptors. More broadly, higher sugar intake can alter microbiota composition, impacting neurotransmitter synthesis and availability [20]. Another dietary aspect linked to alterations in cognitive function and mental health is UP foods. These foods are associated with systemic inflammation, increased oxidative stress, release of advanced glycation end products, and disruption of the endothelial environment, leading to neural damage [21–23].

Individuals with higher intakes of UP foods had about 3 times higher chance of developing hearing loss, a component of the sensory domain. Typically, a higher intake of UP foods is followed by reductions in fresh foods such as fruits and vegetables, a combination that can disrupt the gut microbiome and promote a pro-inflammatory environment, contributing to cochlear infections [24]. Furthermore, a higher salt intake –commonly observed with the intake of UP foods– could also increase hearing loss risk. An imbalance between sodium and potassium in the endolymph can disturb hair cell polarization and the consequent conversion of mechanical sound waves into neural signals [25]. Potassium is mainly found in fruits and vegetables; thus, not surprisingly, diets rich in these foods reduce the chance of hearing loss [26,27]. Regarding vision –the second sensory component– the role of nutrition shows a different perspective. Vision impairment is most often related to food insecurity [28,29], a condition in which food availability, particularly nutrient-rich foods, is disrupted due to insufficient resources and financial concerns [28]. Furthermore, having a vision impairment may lead to nutritional deprivation since difficulties in buying food and preparing meals [28,29]. The impact we have on the environment returns to us, impacting food security. In this sense, the EAT-Lancet diet proposes that food systems should ensure not only human health but also environmental sustainability. Accordingly, this balance can be achieved through widespread, multisectoral, and multilevel action that includes promoting healthy dietary patterns, reducing food waste, and improving food production [30].

The universal reference of a healthy diet consists of providing appropriate energy intake through a higher consumption of a diversity of plant-based foods, and lower amounts of foods from animal sources, saturated-to-unsaturated fat ratio, refined grains, UP foods, and added sugars [30]. These protective diets modulate age-related chronic inflammation (i.e., inflammaging) by reducing inflammatory markers such as C-reactive protein (CRP) and Tumor Necrosis Factor-α (TNF-α),

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in addition to preserving the gut microbiome by preventing or recovering dysbiosis (i.e., microbiota imbalance) and altered gut permeability. Together, these effects prevent suboptimal metabolism, impaired immune and brain function, age-related diseases and conditions, and the decline in overall health associated with aging [31,32].

## Declaration of Generative AI and AI-assisted technologies in the writing process

The authors declare they did not use generative AI and AI-assisted technologies in the writing process.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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